

Conditional Reproduction of Ordinary Matter Fractions

Admissible Targets and Energy Readout in a WRRRA Two Stage Filter Model

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English submission draft r11 8 October 2026

Abstract

We study how a finite model can connect state selection to ordinary-matter and background energy fractions through a common set of operations. The Worldline–Residue–Resource–Action (WRRRA) framework motivates a construction on integer addresses in which separate admission and return filters retain composite relations, while a reversible binary encoding supports a parity-selective readout. With inverse-square address weights and a cutoff of one million, the odd-composite fraction is 4.876893353%. Joint calibration of a common exponent and an admission probability reproduces the benchmark fractions 5%, 26.8%, and 68.2%. We characterize the admissible calibration domain and prove that each finite logistic drive has a unique threshold for any interior target. Different drives can therefore reproduce identical totals while retaining different address distributions. An explicit state-update instrument distinguishes general contractions from the projector case used in the numerical model. A completely positive trace-preserving aggregate channel then transfers the sector probabilities to a declared volume-dependent energy operator. Its pressure derivative yields a conditional homogeneous expansion response with present deceleration parameter -0.523 . The aggregate channel erases the demonstrated distribution differences. The contribution is an executable connection between generation, selection, and energy readout, together with precise limits on what aggregate agreement identifies. The arithmetic encoding, port coupling, and energy operator are specified model choices rather than deductions of a unique microscopic particle theory.

Keywords: finite cosmological model; state selection; common carrier; matter fractions; identifiability; energy readout

1 Introduction

Ordinary matter contributes approximately five percent of the present cosmic energy density. For example, the Planck 2018 baseline flat cosmological model gives $\Omega_b h^2 = 0.0224$ and $h = 0.674$, corresponding to $\Omega_b \approx 0.04930$ [1]. Here (0.05, 0.268, 0.682) is a declared three-sector calibration benchmark at the observational scale. It is not presented as the precise joint best fit of a particular Planck data combination. The denominator is the total energy budget, not the matter-only budget.

The study began with the question: could gravity be understood as the weight of information? Here “weight” motivates an energy-and-pressure load assigned to an information state, rather than a universal conversion from bit count to gravitational mass. This raises a specific modeling problem: information retained without an active matter readout might still carry load. To examine that possibility, we separate what remains, what is expressed as matter, and what contributes to the energy account. The ordinary-matter fraction is therefore investigated as a readout of retained information within this broader question about gravitational sourcing.

Our question is whether one finite construction can generate a candidate population, select what remains, and read the retained and returned sectors through a shared energy account. The Worldline–Residue–Resource–Action (WRRRA) research framework uses minimal computation, a common carrier, and phenotype as organizing principles. In this paper, phenotype means an active matter readout of a retained state; it is not a biological analogy needed for the mathematics. “Upstream” denotes preparation and selection, and “downstream” denotes energy and physical-response readout. We investigate one universe as an execution of the declared rules, without assigning it a privileged mathematical status.

The arithmetic choice makes the composition and reuse of relations explicit. Prime factorization preserves the factors and their multiplicities. The binary quotient and remainder preserve the original address while exposing a readout port. Low-lying zeta zeros supply one concrete phase drive for early admission. These choices make arithmetic relations available to an information-processing interface; they do not identify an integer with a particle, a spatial dimension, or an energy quantum. The model is a prototype for testing the chain from relational structure to specified energy fractions. Its admissible domain and control responses delimit what that chain explains.

The role of a toy model is to isolate such a chain in an inspectable setting. Spekkens illustrates how an explicit restriction on knowledge can organize and test analogues of physical phenomena [2]. Our restriction and target are different: we prescribe address selection and readout, and test a cosmological composition benchmark rather than reconstruct quantum phenomena. We use standard finite-dimensional measurement and channel mathematics [3]. The novelty claimed here lies in the particular connection between address factorization, sequential retention, binary readout, and energy accounting; neither the existence of an isometry nor the measurement formalism is claimed as a new general result.

Four propositions organize the contribution: a lossless realization of the selected port rule; the exact set of admissible joint targets; uniqueness of a finite-drive calibration threshold and consequent non-identifiability from totals; and a conditional aggregate energy response. A central consequence is that differences created during generation can survive in retained address distributions yet disappear completely in the specified energy readout. Reproducing known fractions is a consistency and explanatory result within the declared construction. It does not, by itself, identify the microscopic generator.

2 Shared structure and declared inputs

Minimal computation is used as a design principle: retain a single address account, reuse factor relations, and apply common admission, return, and readout rules. It does not assert a proven global minimum of computational cost. In the constant-admission construction, the address cutoff and classification rules are fixed, and two parameters, α and β , calibrate two targets. The third fraction follows from completeness. In the frame construction, a threshold h replaces β ; the frame count, phase schedule, and frequency list are additional declared inputs. Table 1 separates these roles.

Table 1 Calibration inputs and computed outputs

Quantity	Role	Value or condition
Sector targets	Calibration inputs	$\varphi = 5\%$; $D = 26.8\%$

Quantity	Role	Value or condition
Fixed exponent test	Arithmetic output without the 5% target	$\alpha = 2$; odd composites = 4.876893353%
α and β	Parameters fitted to two targets	$\alpha = 1.8996876951$; $\beta = 0.8654570124$
R and retained total	Outputs of completeness	R = 68.2%; $\varphi + D = 31.8\%$
Frame threshold h	Fitted to the same φ target	h = 1.4476731704; K = 8; J = 4; $\xi = 0.1$
Effective admission	Computed from cumulative address admission	$\beta_{\text{eff}} = 0.8654570124$
Closure responses	Outputs with α and h fixed	K = 4: $\varphi = 3.704795\%$; K = 16: $\varphi = 5.682474\%$

A common carrier means that states share a transport structure while their internal representations and readout effects may differ. The broader program discusses a 15-channel carrier and extensions [4,5]. Those channel counts do not serve as the denominator of the present fractions. This calculation acts on arithmetic addresses; a full address-to-particle map is not assumed. Its implemented downstream connection retains the three sector probabilities through an explicit aggregate channel. It need not retain the full address state. Appendix B relates the terminology to standard mathematical objects.

3 Sequential selection and residual states

3.1 Address relations

The active addresses are the integers 2 through N. A prime address is treated as an independent source; a composite address preserves a multiplicative relation. For example, 77 contains factors 7 and 11, while 49 reuses factor 7 twice. Define the factor count and fold index by

$$n = \prod_p p^{v_p(n)}, \Omega(n) = \sum_p v_p(n), d_{fold}(n) = \Omega(n) - 1, n \geq 2 \quad (1)$$

Thus a prime has fold index zero and a composite has positive fold index. We also use a binary relational dimension: zero for primes and one for composites. It is a classification of address relations, not a measurement of physical spatial dimension. Prime powers remain composite candidates. Address 1, the multiplicative identity, is a separate reference and is excluded from the active normalization. This bookkeeping choice does not assign zero physical vacuum energy to the universe.

3.2 Admission and return

Let ρ be a density operator on the finite address space. Admission A and normal return B are contractions,

$$\rho \geq 0, \text{Tr } \rho = 1, A^\dagger A \leq I, B^\dagger B \leq I, \rho_{in} = A \rho A^\dagger \quad (2)$$

The unnormalized state after admission is ρ_{in} . Combining initial rejection with subsequent normal return gives

$$\begin{aligned} q_R &= 1 - \text{Tr } \rho_{in} + \text{Tr } (B \rho_{in} B^\dagger), \\ q_{res} &= \text{Tr } [\rho A^\dagger (I - B^\dagger B) A], q_R + q_{res} = 1. \end{aligned} \quad (3)$$

The order is retained even when A and B do not commute. In the implemented diagonal model, primes can return normally, whereas admitted composite relations are retained. The motivating picture is an initially broadened admission filter that accepts some composite candidates before the admission window closes, after which the normal return rule remains operative. We call these accepted composites “pseudoprime” candidates only in the WRRRA sense; no primality property of the integer changes, and this is not a Miller–Rabin pseudoprime criterion. The frame dynamics below implement a specified admission response, not a derivation of a physical optical resolution.

3.3 An explicit result instrument

Probabilities alone do not define a post-selection state. One instrument realizing the two-stage probabilities is [3]

$$C = (I - B^\dagger B)^{1/2}, M_0 = (I - A^\dagger A)^{1/2}, M_1 = B A, M_2 = C A. \quad (4)$$

$$\sum_{j=0}^2 M_j^\dagger M_j = I, \tilde{\rho}_{res} = M_2 \rho M_2^\dagger, \rho_{res} = \tilde{\rho}_{res} / q_{res}. \quad (5)$$

The conditional state is defined only when $q_{res} > 0$. Completeness follows by adding $I - A^\dagger A$, $A^\dagger B^\dagger B A$, and $A^\dagger (I - B^\dagger B) A$. Each output is positive and the three probabilities sum to one. Combining outcomes 0 and 1 recovers q_R . This instrument is an explicit realization, not a unique physical apparatus.

For a general contraction B, a residual result need not lie in $\ker B$. With $A = I$ and $B = I/2$, $q_{res} = 3/4$ although $\ker B = \{0\}$. In the implemented projector case $B = P_{\text{prime}}$, C is the composite projector P_C , and the conditional state is $P_C A \rho A^\dagger P_C / q_{res}$. It is supported in the return kernel.

$$K = \ker B, \text{supp } \rho_{res} \subseteq K, U_k K \subseteq K \quad (6)$$

In this projector implementation, assume subsequent updates U_k are unitary and satisfy the displayed invariant-subspace condition. Then trace and positivity are preserved and all subsequent return probabilities vanish. This is a sufficient condition for permanent return blocking, not a necessary condition for every long-lived or metastable residue. Leakage outside the kernel requires a new lifetime and fraction calculation. Initial filter closure alone does not establish the invariance assumption.

4 A fixed-weight benchmark

For $\alpha > 1$, normalize the finite address weights as

$$Z_N(\alpha) = \sum_{n=2}^N n^{-\alpha}, w_n = \frac{n^{-\alpha}}{Z_N(\alpha)}, n^{-\alpha} = \prod_p p^{-\alpha v_p(n)} \quad (7)$$

The product form reuses factor contributions. The Euler product is the standard number-theoretic context for these weights [6]. The finite sums also exist for other exponents, but $\alpha > 1$ is the adopted family here. The first test fixes $\alpha = 2$ and uses the smallest prime to separate odd composites. A finite phase selector and the selected weight are

$$H_p(n) = 1 - \frac{1}{p} \sum_{r=0}^{p-1} e^{2\pi i r n / p} = 1 - 1_{p|n}, f_{odd}(N, \alpha) = \sum_{\substack{2 \leq n \leq N \\ n \notin P, 2 \nmid n}} w_n \quad (8)$$

Here $\mathbb{P}$ denotes primes. At $N = 1,000,000$, the odd-composite weight is 0.04876893353364531. The 5% target is not inserted into this fixed-exponent calculation. Of its 4.876893353 percentage points, 2.498328130 come from prime powers and 2.378565223 from composites having at least two distinct prime factors. Table 2 gives the cutoff dependence. This arithmetic output and the next section's calibrated output are separate results.

Table 2 Odd-composite weight with α fixed at 2

Cutoff N	Odd-composite fraction
161	4.600993937%
10,000	4.871477356%
1,000,000	4.876893353%
Analytic comparison limit	4.876952810%

$$P(\alpha) = \sum_{p \in \mathbb{P}} p^{-\alpha}, f_{odd, \infty}(\alpha) = \frac{(1 - 2^{-\alpha})(\zeta(\alpha) - 1) - P(\alpha)}{\zeta(\alpha) - 1}, \alpha > 1 \quad (9)$$

$P(\alpha)$ is the prime zeta sum and $\zeta(\alpha)$ is the Riemann zeta function. The expression subtracts the prime contribution from the odd active weight, giving 4.876952810% at $\alpha = 2$. The limit is an analytic comparison for truncation error, not an assumption of infinitely many physically realized addresses. A separate uniform-weight sieve on addresses 2 through 161, using divisibility by 2, 3, and 5, leaves $8/160 = 5\%$; at cutoff 321 its fraction is 6.875%. That finite-window coincidence is not used as the joint model.

5 Joint fractions and a reversible readout

5.1 The selected port rule

Write C_e and C_o for even and odd composites. Even composites enter with unit probability and form sector D ; odd composites enter with probability β and form the active readout sector ϕ . Primes return normally. The diagonal filters are

$$\begin{aligned} A_{nn} &= 1_p(n) + 1_{C_e}(n) + \sqrt{\beta} 1_{C_o}(n), B_{nn} = 1_p(n) \quad (10) \\ E_D(n) &= 1_{C_e}(n), E_\phi(n) = \beta 1_{C_o}(n), \\ E_R(n) &= 1_p(n) + (1 - \beta) 1_{C_o}(n), \quad (11) \\ E_D(n) + E_\phi(n) + E_R(n) &= 1, f_s = \sum_{n=2}^N w_n E_s(n). \end{aligned}$$

For $0 \leq \beta \leq 1$, the three positive effects sum to the identity, so they form a positive operator-valued measure. Their probabilities partition any normalized input. Our numerical input is diagonal with entries w_n ; the operator notation also states the realization and state-update conditions clearly. The numerical example does not require nonclassical coherence.

The final readout reuses binary grouping. Write the address as a quotient and one remainder bit,

$$n = 2m + r(n), m = \text{floor}(n/2), r(n) = n - 2m \in \{0, 1\}. \quad (12)$$

$$V|n\rangle = |m\rangle \otimes |r(n)\rangle, V^\dagger V = I, n = 2m + r(n). \quad (13)$$

To avoid a later notational ambiguity, V in these three port equations denotes an isometry; the scalar V in Section 7 denotes physical volume. The pair (m,r) uniquely reconstructs n . A remainder bit equal to one is called the unpaired port. This is an encoding of an integer, not a count of physical bits or a claim that the integer equals energy. Let P_C project onto all composite addresses. Pull back the port effect and restrict it to the retained space:

$$\begin{aligned} \Pi_1 &= V^\dagger (I \otimes |1\rangle\langle 1|) V, \\ Q_\phi &= P_C \Pi_1, Q_D = P_C (I - \Pi_1), Q_\phi + Q_D = P_C. \end{aligned} \quad (14)$$

$$\begin{aligned} E_\phi &= A^\dagger Q_\phi A, E_D = A^\dagger Q_D A, \\ E_R &= I - A^\dagger P_C A, \\ E_\phi(n) &= \beta 1_C(n) r(n), E_D(n) = 1_C(n) (1 - r(n)), \\ E_\phi + E_D + E_R &= I. \end{aligned} \quad (15)$$

The diagonal operators commute in this realization. Hence Q_ϕ and Q_D are orthogonal projectors with sum P_C , and the pulled-back effects reproduce the previous sector values. Composite 8 has remainder zero, composite 9 has remainder one, and prime 7 has remainder one but is excluded by P_C . The port value one is distinct from the reference address 1. β remains an admission probability, not an additional fitted port efficiency.

Proposition 1. For the declared finite address basis, prime return projector, admission operator, and selection of port $|1\rangle$, the binary map is an isometry and its pulled-back effects realize the original three-sector partition without a new calibration parameter. *Proof.* Distinct n give distinct orthonormal pairs (m,r) , hence $V^\dagger V = I$. Substituting $r(n) = 1_{\text{odd}}(n)$ into the displayed effects gives the sector partition. This also preserves the original factorization because n is exactly recoverable.

The proposition establishes consistency of the chosen coupling, not its unique physical necessity. Any finite partition can be encoded with labels. What is reused here is a simple quotient–remainder representation alongside the existing factor relation and filters. A microscopic carrier obeying this port rule remains to be specified. `render_coupling_check.py` checks the finite realization and the unchanged benchmark through 20 implementation tests [7].

5.2 Joint calibration

Every even composite is $2m$ with $m \geq 2$. Therefore

$$f_2(N, \alpha) = 2^{-\alpha} \frac{Z_{\lfloor N/2 \rfloor}(\alpha)}{Z_N(\alpha)} \rightarrow 2^{-\alpha} (\alpha > 1) \quad (16)$$

First solve $f_2(N, \alpha) = 0.268$ at $N = 1,000,000$. At that exponent the available odd-composite fraction is 0.0577729445632. Then admit the proportion required for the 0.05 target:

$$\alpha = 1.899687695055436, \beta = \frac{0.05}{0.057772944563189425} = 0.865457012413696 \quad (17)$$

The returned fraction is the complement. Table 3 reports the common normalization. This is a two-target calibration followed by a third-sector output, not three independently fitted sectors. Calibration using

measured scales is a legitimate model construction; disclosing it allows readers to assess which parts of the result follow from shared structure.

Table 3 Joint partition of the full active address weight

Sector	Fraction	Interpretation
φ	5.0000000000%	Retained with active matter readout
D	26.8000000000%	Retained without active matter readout
Retained total	31.8000000000%	$\varphi + D$
R	68.2000000000%	Initial rejection plus normal return

Keeping α and β fixed while changing the cutoff gives Table 4. In particular, the 4.8769% result at $\alpha = 2$ must not be added to an even-composite result calculated at another exponent. All sectors of the joint construction use the same denominator and exponent.

Table 4 Cutoff response with α and β fixed

N	φ	D	R
10,000	4.989044%	26.791475%	68.219480%
100,000	4.998716%	26.799046%	68.202237%
1,000,000	5.000000%	26.800000%	68.200000%

5.3 The admissible target domain

Proposition 2. Fix $N \geq 9$ and adopt $\alpha > 1$. Let $d_N(\alpha) = f_2(N, \alpha)$ and $o_N(\alpha) = f_{\text{odd}}(N, \alpha)$. The exact set of jointly realizable targets is

$$D_N = \{(\phi, d, r) : \exists \alpha > 1, d = d_N(\alpha), 0 \leq \phi \leq o_N(\alpha), r = 1 - d - \phi\}. \quad (18)$$

Proof. The effects give $d = d_N(\alpha)$ and $\varphi = \beta o_N(\alpha)$, with $0 \leq \beta \leq 1$, establishing necessity. Conversely $o_N(\alpha) > 0$ for $N \geq 9$. For any point in the displayed set, $\beta = \varphi/o_N(\alpha)$ lies in $[0, 1]$ and realizes all three targets. No global uniqueness of the inverse of finite d_N is needed.

At the calibrated exponent, φ ranges from zero to approximately 5.777294456%. A target $\varphi = 6\%$ at the same α and d would require $\beta = 1.038548415$ and is inadmissible. Changing d requires reconsidering α and the available weight. In the analytic comparison limit, $d = 2^{-(\alpha)}$ gives $0 < d < 1/2$ and $\alpha = -\log_2 d$. The finite model remains the executed case.

6 Finite drives and what totals cannot identify

6.1 Frame admission

A frame-resolved version replaces constant β by a cumulative admission probability. Let γ_j be the positive imaginary parts of the selected low-lying zeta zeros. Define a finite real drive and logistic admission response by

$$\delta_n(k) = \sum_{j=1}^J c_{jk} \cos(\gamma_j \log n + \phi_{jk}), a_n(k) = g(\delta_n(k)), 0 \leq g \leq 1 \quad (19)$$

$$\begin{aligned} a_n(k) &= \frac{1}{1 + \exp(h - \delta_n(k))}, 0 \leq k < K, \\ T_n(K) &= 1 - \prod_{k=0}^{K-1} (1 - a_n(k)). \end{aligned} \quad (20)$$

The baseline uses $J = 4$, $c_{jk} = 1/\sqrt{J}$, phase $\phi_{jk} = \gamma_j \xi k$, $\xi = 0.1$, and $K = 8$ [7]. The frames and phase advance are dimensionless. A physical time per frame is not supplied. The previous calculation checked the first eight zeros at two arithmetic precisions and with an Euler–Maclaurin cross-check; the baseline drive uses four. No hypothesis about all zeta zeros is required.

For odd composites, the initial unadmitted weight is w_n . Each frame transfers $a_n(k)$ of the remaining weight to the retained account; admitted weight is not offered again. At $k \geq K$ no new admission occurs. Even composites enter at the first frame under the prescribed rule. This produces

$$\begin{aligned} f_\phi(K) &= \sum_{n \in C_o} w_n T_n(K), \beta_{eff}(K) = \frac{f_\phi(K)}{\sum_{n \in C_o} w_n}, \\ f_R(K) &= 1 - f_D - f_\phi(K). \end{aligned} \quad (21)$$

With the joint α fixed, $h = 1.4476731704$ gives the 5% baseline. β_{eff} summarizes the result and is not a second admission input. Holding α and h fixed while closing after four or sixteen frames gives Table 5. These are conditional scenario outputs; changing the physical interpretation of K would require a time map.

Table 5 Closure response at fixed threshold

Closure K	ϕ	D	R
4	3.704795%	26.800000%	69.495205%
8	5.000000%	26.800000%	68.200000%
16	5.682474%	26.800000%	67.517526%

6.2 Equal totals and different distributions

We compare the baseline drive with a static phase using the same zeros and $\xi = 0$, equally spaced frequencies (14,21,28,35) with $\xi = 0.1$, and a constant drive $\delta_n(k) = 0$. All use the same address weights, $K = 8$, depletion rule, and final readout. At the common threshold their totals differ as in Table 6.

Table 6 Drive comparison at a common threshold

Drive	Common h	ϕ	R
Zeta phase	1.447673	5.000000%	68.200000%
Static zeta phase	1.447673	4.859796%	68.340204%
Equal frequencies	1.447673	5.095593%	68.104407%
Constant drive	1.447673	4.710550%	68.489450%

Recalibrating each threshold to the same 5% target yields identical three-sector totals but different address distributions. Let π_n be the normalized distribution within the retained ϕ sector. Relative to baseline $x = 0$, define

$$\pi_n^{(x)} = \frac{w_n T_n^{(x)}}{\sum_{m \in C_o} w_m T_m^{(x)}}, \quad (22)$$

$$\Delta_A^{(x)} = \frac{\sum_{n \in C_o} w_n |T_n^{(x)} - T_n^{(0)}|}{\sum_{n \in C_o} w_n}, \quad TV^{(x)} = \frac{1}{2} \sum_{n \in C_o} |\pi_n^{(x)} - \pi_n^{(0)}|.$$

Δ_A compares admission probabilities using available odd-composite weights; total variation TV compares the normalized retained distributions. The two statistics have different denominators. Table 7 shows the threshold, admission at address 9, and TV after recalibration.

Table 7 Address differences after matching the same total

Drive	Fitted h	Admission at 9	TV
Zeta phase	1.447673	85.681301%	0.000000%
Static zeta phase	1.327223	80.386717%	5.163822%
Equal frequencies	1.525841	88.580085%	1.564832%
Constant drive	1.255378	86.545701%	0.721800%

A second readout groups addresses by their smallest prime factor: first factor 3, then 5 and 7 among the remaining addresses, and finally all others. Table 8 records each family's contribution to the full address weight. Rows sum to 5% up to rounding. Thus this distinction is calculated within the retained state; it is not yet a measured particle-family distribution.

Table 8 Family contributions at the same 5% total

Drive	Factor 3	Factor 5	Factor 7	Other
Zeta phase	4.018338%	0.666538%	0.199781%	0.115343%
Static zeta phase	4.020562%	0.663793%	0.202863%	0.112781%
Equal frequencies	4.037629%	0.650211%	0.198701%	0.113459%
Constant drive	4.023100%	0.662494%	0.198626%	0.115780%

phase_control_comparison.py reproduces the controls and checks normalization, family decomposition, and the prior numerical records through 36 tests [7]. Equal totals do not select the zeta drive as uniquely physical. They coexist with computable differences in the generated state.

6.3 Threshold uniqueness and non-identifiability

Proposition 3. Fix finitely many odd-composite addresses with positive weights, finite $K \geq 1$, and finite real drives $\delta_n(k)$ independent of h . Then $\Phi(h) = \sum_n w_n T_n(h)$ is continuous and strictly decreasing, with limits o_N as h tends to $-\infty$ and zero as h tends to $+\infty$. Every target $0 < \varphi^* < o_N$ has exactly one threshold.

$$\frac{dT_n}{dh} = - \left[\prod_{k=0}^{K-1} (1 - a_n(k)) \right] \sum_{k=0}^{K-1} a_n(k) < 0. \quad (23)$$

Proof. The logistic derivative is $da_n/dh = -a_n(1 - a_n)$. Differentiating $1 - T_n = \prod_k (1 - a_n)$ gives the displayed strictly negative derivative for finite h . Positive weighted summation preserves strict decrease. The endpoint limits and intermediate value theorem establish existence and uniqueness. The exact endpoints are approached only in the limits, not at finite thresholds.

Consequently, distinct finite drives can each be calibrated to the same interior total. The control distributions establish that the remaining freedom need not be empty relabeling: address 9 and the family readouts distinguish the fitted drives. By contrast, the aggregate channel in the next section assigns the same output to any inputs with the same three totals. Homogeneous expansion computed through that channel cannot discriminate these generators.

7 Energy readout and a conditional expansion response

7.1 From address weight to energy fraction

An address probability is not automatically an energy fraction. For nonnegative diagonal address loads e_n and a positive total load, define

$$F_s^{(E)} = \frac{\sum_n w_n e_n E_s(n)}{\sum_n w_n e_n}, e_n \geq 0, \sum_n w_n e_n > 0 \quad (24)$$

This definition uses the same sector effects with a new, explicitly stated energy weighting. Equal nonzero loads give $F_s^{(E)} = f_s$. Independence of prime addresses does not force equal loads. If only even-composite addresses carry twice the baseline load, the calibrated address state yields energy fractions

(0.05,0.536,0.682)/1.268, or approximately (3.943218%,42.271293%,53.785489%). This is a response to a changed physical readout, not an error in the equal-load benchmark.

The retained total 31.8% uses the full active normalization. Within the retained sector alone, the active readout fraction is $5/31.8 \approx 15.72\%$; it is not the full-universe 5%. Nor is it a gravitational mass fraction until the energy map is supplied. “Information load” here denotes the assigned contribution to the energy and pressure account. An inactive matter readout need not imply zero energy.

7.2 An explicit aggregate channel

Use three orthonormal outcome labels $|\phi\rangle$, $|D\rangle$, and $|R\rangle$ and define the measure-and-prepare channel [3]

$$F_{agg}(\rho) = \sum_{s \in \{\phi, D, R\}} \text{Tr}(\rho E_s) |s\rangle\langle s|. \quad (25)$$

It is completely positive and trace preserving. Indeed, for a spectral decomposition $E_s = \sum_j \lambda_{sj} |v_{sj}\rangle\langle v_{sj}|$, Kraus operators $K_{sj} = \sqrt{\lambda_{sj}} |s\rangle\langle v_{sj}|$ satisfy $\sum_{sj} K_{sj}^\dagger K_{sj} = I$ and give the displayed map. The channel preserves sector probabilities, while generally discarding address distributions and coherences. It is an implemented coarse readout rather than a reversible microscopic particle map.

To compare drives through one and the same downstream channel, the distinction between selection and coarse readout must be explicit. For drive x , replace the odd-composite effect by $E_\phi^{(x)}(n) = T_n^{(x)} 1_{(C_o)}(n)$, keep $E_D^{(x)} = 1_{(C_e)}$, and set $E_R^{(x)} = 1 - E_\phi^{(x)} - E_D^{(x)}$. With the same input weights, define the state that records both outcome s and address n :

$$\omega_x = \sum_{n,s} w_n E_s^{(x)}(n) |s,n\rangle\langle s,n|, C(\omega_x) = \text{Tr}_n \omega_x = \sum_s f_s^{(x)} |s\rangle\langle s|. \quad (26)$$

This state is positive with trace one. Its conditional ϕ -address distribution is exactly $\pi_n^{(x)}$ used in the control comparison. The preparation of ω_x depends on the drive, whereas the partial trace C is identical for every drive. Equal fitted totals therefore give identical outputs of C even when the recorded address distributions differ. The preceding measure-and-prepare formula abbreviates this composition for a fixed set of effects; comparing drives changes those effects at preparation, not the final partial trace. In particular, for two fitted drives with the same totals,

$$C(\omega_x - \omega_y) = 0, \text{Tr}[C(\omega_x) H_{agg}(V)] = \text{Tr}[C(\omega_y) H_{agg}(V)]. \quad (27)$$

No observable on the three-label output, including the assigned energy, distinguishes this pair. This conclusion follows from an explicit common information-discarding operation; it is not a claim that the same input state was transformed differently by one fixed upstream channel.

A fixed address-sensitive load supplies a contrasting readout before aggregation. Set $e_9 = 2$ and all other $e_n = 1$ in relative units, identically for all four calibrated drives. Apply this load to ω_x while the address label is still available. Since $w_9 = 0.0205169848463$, the energy fractions are

$$F_\phi^{(x)} = \frac{0.05 + w_9 T_9^{(x)}}{1 + w_9}, F_D^{(x)} = \frac{0.268}{1 + w_9}, F_R^{(x)} = \frac{0.682 + w_9 (1 - T_9^{(x)})}{1 + w_9}. \quad (28)$$

In the order zeta phase, static phase, equal frequencies, and constant drive, F_ϕ is respectively 6.622057%, 6.515612%, 6.680336%, and 6.639436%; F_D is 26.261199% in every case and F_R is the complement. The same specified readout therefore distinguishes the generated states. It cannot be

implemented from the three-label output alone, which has discarded the address. Address 9 is chosen because its admission already appears in Table 7, not because a measured particle energy has been assigned to it. This is a conditional diagnostic of readout resolution, not an additional fit or observational claim.

Let $u_* > 0$ have units of energy density and $V_0 > 0$ be a reference volume. Assign

$$H_{agg}(V) = u_* \text{diag}(V_0, V_0, V), E(V) = \text{Tr}[F_{agg}(\rho) H_{agg}(V)]. \quad (29)$$

$$E(V) = u_* \left[(f_\phi + f_D) V_0 + f_R V \right], p = - \left. \frac{\partial E}{\partial V} \right|_{f_s, u_*, V_0} = -u_* f_R. \quad (30)$$

Proposition 4. Under this declared energy operator, the energy fractions at $V = V_0$ equal the sector probabilities. The first two sectors have volume-independent total energy and hence zero pressure; the R sector has energy proportional to volume and $p_R/\epsilon_R = -1$ when $f_R > 0$. For each sector and their sum, $dE = -p dV$ holds at fixed fractions, u_* , and V_0 . Proof. Evaluate the diagonal trace, divide the three energies by the total at V_0 , and differentiate with respect to V .

This is a conditional realization of matter-like and background-like responses. The equations of state follow from the chosen volume dependence, not from prime classification alone. At other volumes the instantaneous energy fractions change; the f_s in the operator trace remain reference probabilities. The pressure is a uniform volume-response pressure. A general local stress tensor is not derived.

7.3 Homogeneous response

Set the present scale factor to $a = 1$ and $V = V_0 a^3$. Adopting a spatially flat homogeneous Friedmann response, neglecting radiation over the intended late-time regime, and identifying u_* with the present critical energy density gives [8]

Here $H = \dot{a}/a$ is the Hubble expansion rate, H_0 its present value, ϵ the total energy density, and q_0 the present value of $-a \ddot{a}/\dot{a}^2$. The scale factor a and the reference sector probabilities are dimensionless.

$$V = V_0 a^3, \epsilon(a)/u_* = (f_\phi + f_D) a^{-3} + f_R, H^2/H_0^2 = (f_\phi + f_D) a^{-3} + f_R. \quad (31)$$

$$q_0 = 1/2 (f_\phi + f_D) - f_R = 1/2 (1 - 3f_R) = -0.523. \quad (32)$$

The expansion formula is evaluated within the supplied downstream computation using the upstream totals and a uniform downstream carrier state I/N_C [9]. N_C is a downstream lattice size, distinct from the address cutoff N . This is an explicit aggregate substitution, not transmission of the entire microscopic address state. Because the totals and equations of state fix q_0 , its reproduction checks the connection; it is not a further independent datum fitted in this paper.

The prior energy-reference computation tests the same final $E(V)$ and its derivative in a four-state representation with subtraction of a common reference term and addition of a background term. `vacuum_reference_check.py` passed 33 checks; `twist_bridge_check.py` passed 12 checks of the conditional downstream connection; `benchmark_derivation.py` passed six checks of the finite sums, calibration, and deceleration coefficient. Original downstream calibration values remain preserved. The fraction substitution discussed here gives -0.523 ; it does not overwrite a differently calibrated earlier scenario. The present paper does not perform a new supernova or expansion-data likelihood fit.

8 Verification and conditions of failure

The evaluation sequence is verified inputs, WRR transformation, outputs, and conditions of failure. Inputs include factorization, the active cutoff, the fixed test exponent or stated calibration targets, drive parameters, and the energy-readout choices. The transformation applies admission, return, parity readout, and the specified aggregate energy map. Outputs include the fractions, admissible target domain, address-control differences, energy derivative, and homogeneous response. Table 9 distinguishes mathematical checks from physical implementation and empirical tests.

Table 9 Claims and their corresponding failure conditions

Claim	Failure condition	Level
Partition and update	Negative probabilities, incomplete effects, or failed reproduction	Mathematics and implementation
Permanent return blocking	Support or invariant-update assumptions fail	Declared dynamics
Physical port coupling	A specified carrier fails to obey the selected coupling	Physical implementation
Aggregate energy response	Sector energies, pressure derivative, or work identity disagree for the same $E(V)$	Conditional calculation
Empirical correspondence	Fixed carrier, load, clock, and observation maps produce results incompatible with data	Observation after maps are specified

The additional `r10_check.py` suite passes 27 representative checks: noncommuting contraction updates, the projector special case and a leaking-update counterexample, feasible and infeasible targets, the logistic derivative and calibration, aggregate-channel completeness and information loss, nonuniform loads, and the pressure derivative. Five independently rerun earlier suites provide 107 passing implementation checks. Counts refer to software assertions, not independent observational confirmations. The general propositions are established by their proofs, not by extrapolating numerical samples.

The revision adds eight checks of recorded outcome—address states and their common partial trace, and nine checks reproducing address-9 admission and the address-sensitive energy readout. Both suites pass. These 17 checks test the explicit connection introduced during review; they do not increase the count of empirical confirmations.

Strict monotonicity is sampled where floating-point differences remain resolvable; near saturation finite precision can round different totals to the same value. The analytic derivative establishes the real-variable statement. The code, numerical outputs, dependency versions, and previous execution records are supplied so that this distinction can be audited.

For an empirical test, the physical carrier, energy loads, time conversion, and observation map must first be fixed. A disagreement exceeding specified observational and numerical uncertainties would reject that particular implementation. Revising those inputs defines a revised model to be retested; it does not retroactively count as passage by the old setting. The currently demonstrated arithmetic and aggregate calculations do not supply a missing microscopic observation map by implication.

9 Discussion and conclusion

The construction reproduces an ordinary-matter-scale fraction with fixed inverse-square weights and jointly realizes the three benchmark sectors after a transparent two-parameter calibration. Its contribution is the common mechanism connecting these calculations: a finite relational state, distinct admission and return, a reversible address encoding with selected port coupling, and an explicitly executed energy readout. Calibration does not erase this structural content, but it determines what the numerical agreement alone can establish.

The admissible domain provides a concrete limitation: at the fitted exponent the active fraction cannot exceed approximately 5.7773%. The drive comparison and uniqueness proposition identify a second limitation: matching the total cannot select the generator. The aggregate channel makes this limitation operational. It discards precisely the address information on which the fitted controls differ, so identical homogeneous energy responses coexist with different generation histories. A more detailed physical readout would need to preserve and measure some of those differences.

Relative to standard measurement theory [3], this is a specified use of effects and instruments rather than a new measurement postulate. Relative to informational toy theories such as [2], its target is a selection-to-energy interface rather than a reconstruction of quantum phenomena. Relative to homogeneous cosmology [8], it supplies a declared address account for reference fractions and evaluates an adopted response. These distinctions locate the explanatory contribution without assigning to the arithmetic encoding a derivation it does not perform.

The calculation returns to the motivating question in a limited but explicit way: inactive matter readout and nonzero energy load are compatible within one account, and the same assigned energy produces a pressure response. This establishes an executable version of the proposed distinction between expression and load. It does not establish that physical gravity universally measures information, nor derive a unique load from an information count. The gravitational response used here is the declared homogeneous connection.

If the fixed rules violate completeness, the residual state update, or the pressure derivative of the same energy function, the proposed implementation fails. If a subsequently fixed physical readout conflicts with measurements, that realization must be revised or rejected. Conversely, reproduction of known values under the stated rules is a realized consistency result. The next substantive task is to specify a carrier and readout that connect retained address differences to particle or energy observables, while preserving the same accounting conditions.

Appendix A Numerical details and sensitivity

For independent reproduction, compute the prime sum and the even-composite identity on the same address range:

$$P_N(\alpha) = \sum_{\substack{p \leq N \\ p \in P}} p^{-\alpha},$$

$$f_2(N, \alpha) = 2^{-\alpha} \frac{Z_{N/2}(\alpha)}{Z_N(\alpha)}, \quad (33)$$

$$f_{odd}(N, \alpha) = 1 - f_2(N, \alpha) - \frac{P_N(\alpha)}{Z_N(\alpha)}.$$

$$\begin{aligned} Z_N(2) &\approx 0.644933066848726, P_N(2) \approx 0.452247352265374, \\ Z_{N/2}(2) &\approx 0.644932066850226, \\ f_{odd}(N, 2) &\approx 1 - 0.249999612363456 - 0.701231454102898, \\ f_{odd}(N, 2) &\approx 0.048768933533645. \end{aligned} \quad (34)$$

The analytic comparison uses $\zeta(2) - 1 \approx 0.644934066848227$ and $P(2) \approx 0.45224742004106$. The standard Möbius inversion of the logarithmic Euler product gives $P(\alpha) = \sum_{k \geq 1} \mu(k) \log \zeta(k\alpha)/k$ for $\alpha > 1$ [6]. The finite implementation uses a prime sieve and finite sums; it does not depend on a physically infinite construction.

$$\begin{aligned} f_2(N, \alpha) &= 0.268 \Rightarrow \alpha \approx 1.8996876950554, \\ f_{odd}(N, \alpha) &\approx 0.0577729445632, \\ \beta &= \frac{0.05}{f_{odd}(N, \alpha)} \approx 0.8654570124137, \\ (f_\phi, f_D, f_R) &= (\beta f_{odd}, f_2, 1 - f_\phi - f_D) = (0.05, 0.268, 0.682). \end{aligned} \quad (35)$$

The common exponent is solved in the bracket [1.8, 2.0]. At the solution, $Z_N \approx 0.750107847555495$, $Z_{N/2} \approx 0.750104000487848$, and $P_N \approx 0.505743005317387$. The script path from the reproducibility root is `review_r8/benchmark_derivation.py`. Values displayed here are rounded; executing the scripts reproduces their own recorded floating-point values within the stated tolerances.

Table 10 changes α by ± 0.01 while holding β fixed. This is a parameter response, not an observational uncertainty estimate. Changing β by 0.01 at the fixed exponent instead changes ϕ by approximately 0.057773 percentage points, with the opposite change in R and no change in D. Probabilities outside [0,1] are not admissible.

Table 10 Exponent sensitivity with admission probability fixed

α	ϕ	D	R
1.889687695	5.086202%	26.986391%	67.927407%
1.899687695	5.000000%	26.800000%	68.200000%
1.909687695	4.915451%	26.614894%	68.469655%

Appendix B Terminology and scope

A common carrier is a shared transport structure. The larger research program uses a decomposition into channels,

$$H_C = \bigoplus_{i=1}^{15} H_i, D_C: H_C \rightarrow H_C \quad (36)$$

This carrier space is not identified with the finite address space merely by writing the same state symbol. A general address-to-physical-state map must be specified before particle readout. Its schematic form is

$$\sigma = F(\rho), \sigma \geq 0, \text{Tr } \sigma = 1, P_\phi = \text{Tr}(\sigma Q_\phi), E = \text{Tr}(\sigma H) \quad (37)$$

In the implemented example, the map is the three-label aggregate channel and the energy operator is $H_{\text{agg}}(V)$. The 15-channel context and its proposed extensions [4,5] are background to the common-carrier design; they are not invoked as a completed microscopic integration in the present calculation. Table 11 gives a compact vocabulary for readers from other fields.

Table 11 Terms and their mathematical roles

Term	Standard description	Role in this paper
Upstream / downstream	State preparation and selection / physical readout	Successive computational stages
Common carrier	Shared transport structure	A reuse principle; microscopic map specified separately
Dimension filters A, B	Contraction operators	Admission and normal return
Retained sector	Post-selected residual state	$\phi + D = 31.8\%$ of the full input
Phenotype ϕ	Readout-active retained sector	Admitted odd composites under the selected rule
Unexpressed sector D	Readout-inactive retained sector	Even composites; energy need not vanish
Return R	Rejection and normal return	Complementary effect
Fold and relational dimension	Factor multiplicity and a binary relation class	Not spatial dimension
Renderer and unpaired port	Isometry and parity-selective readout	Reversible address encoding, declared coupling
Reference address 1	Multiplicative identity	Excluded from active weights; distinct from R

The symbol K in frame formulas is a positive integer, whereas $K = \ker B$ in the retention condition is a subspace. P_C is the composite projector, $P(\alpha)$ is the prime zeta sum, and p is pressure. The scalar volume V and the port isometry V occur in different sections; neither is a new free calibration parameter. Address-level energy e_n and the aggregate energy operator specify different resolutions of the readout. The numerical probabilities are not identified with physical particle abundances without those choices.

Statements and Declarations

Data and code availability. The earlier Korean r9 manuscript is publicly archived at <https://doi.org/10.5281/zenodo.23112865>. The underlying upstream calculations are archived in [7], and the downstream energy reference in [9]. The accompanying revision package contains the present editable

manuscript, finite-sum and control code, r10_check.py, numerical outputs, and a reproducibility README. No newly measured observational dataset is reported. The present English revision has not yet been assigned a separate public DOI.

Author contributions. Wonsik Choi and Jeongin Choi contributed to the conceptual construction, interpretation, and manuscript review in the preceding version. Wonsik Choi organized the computational materials and led manuscript preparation. Both authors have read and approved the final manuscript and consent to its submission.

AI assistance. ChatGPT assisted with formalization, code drafting, numerical checks, translation, and editing. An additional AI review assessed mathematical consistency and claim scope. This assistance is disclosed as part of the preparation method and is not external journal peer review. The named human authors retain responsibility for the submitted work.

Funding and competing interests. Author declarations are to be supplied before submission; no funding or absence-of-conflict statement is inferred from the available research record.

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